Miranda Caputo

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SUMMARY

Ph.D. candidate in Physics with 6+ years of experience building scalable data pipelines, statistical models, and automated analysis systems for large imaging-based datasets. Proven ability to design end-to-end data workflows, perform hypothesis-driven analysis, extract actionable insights from complex data, and publish results in respected, peer-reviewed journals. Seeking Data Scientist role within AIML.

SKILLS

- *Languages:* Python (7+ years), C++, SQL
- *Data Science & Analytics:* Statistical modeling, hypothesis testing, data analysis, experimentation, feature engineering, model validation, data wrangling
- *Machine Learning & AI:* Retrieval-Augmented Generation (RAG), prompt engineering and design, LLM evaluation, classification models
- *Data Engineering:* Scalable data pipeline development, ETL workflows, data cleaning, structured dataset generation
- *Libraries & Tools:* Pandas, NumPy, SciPy, Jupyter, Git
- *LLM & AI Tools:* OpenAI API, LLM integration, prompt optimization
- *Visualization:* Matplotlib, exploratory data analysis, scientific visualization
- *Systems & Environments:* MacOS, Linux/Unix, Bash

EDUCATION

Ph.D. in Physics, Conc. Astrophysics, University of Toledo — **2020 - July 2026 (expected)**
Thesis Advisor: *Prof. Rupali Chandar* — Toledo, OH
Untangling the Formation and Dissolution of Optically Bright and Obscured Stellar Clusters in Nearby Extreme Environments

B.Sc. in Physics, Santa Clara University — **2014 - 2018**
Santa Clara, CA

EXPERIENCE

Graduate Research Assistant, University of Toledo — **2020 - present**
Advisor: *Prof. Rupali Chandar* — Toledo, OH

- **Led development of scalable, end-to-end Python data pipelines to process GB-scale datasets, transforming raw data into structured datasets for statistical modeling and analysis of 10,000+ samples**
  - Designed and implemented a **retrieval-augmented generation (RAG) pipeline using LLMs (OpenAI APIs)** to classify and validate feature extraction outputs, improving accuracy and robustness in large, noisy datasets
  - Developed and iterated on prompt engineering strategies to improve LLM performance, and conducted systematic LLM evaluation to assess accuracy, consistency, and reliability of model outputs vs human visual classification
  - Utilized SQL to query, and join structured datasets, improving reliability of downstream statistical modeling and reducing manual data inputs by 10x
  - Developed production-quality pipelines and code used in 4 published and 3 ongoing research projects, enabling reproducible analysis and international collaboration across teams
- **Image manipulation for data visualization**
  - Developed data visualization workflows to analyze and communicate patterns in high-dimensional datasets, improving interpretability and processing efficiency
  - Visualizations used in 8+ peer-reviewed publications, including featured scientific highlight on AAS Nova

Leadership & Collaboration

- **Collaborated with scientists across North America** on peer reviewed papers: six published, one submitted, and six forthcoming with two first-author papers published and two in progress
- Presented **research talks at an international conference and a major national science conference** along with multiple poster presentations and community science talks
- Mentored 20+ students in data analysis and Python workflows, improving team productivity and onboarding, and guiding research direction
- Created Python tutorials to teach image manipulation to graduate and undergraduate students, **leading to multiple published images**

Research Assistant, NASA Ames, SOFIA Science Center/USRA — **2019 - 2020**
Advisor: *Dr. B-G Andersson* — Mountain View, CA

- **Built Python-based data pipelines to process and analyze large-scale observational datasets**, enabling statistical analysis of complex polarization patterns during flight

GRANTS AND AWARDS

JWST Cycle 3: (P.I.) GO-06035 — **Feb. 2025**
Space Telescope Science Institute

- **Principal Investigator for highly competitive James Webb Space Telescope program (11% acceptance rate)**, leading full lifecycle data analysis project

Astronomy Seminar Best Graduate Student Presentation — **Spring 2022 & Spring 2024**
University of Toledo

- Winner of the best research talk in two separate semesters covering topics outside field of expertise (2020 Astronomy Decadeal Survey and habitable exoplanets).

Clare Boothe Luce Scholar — **Summer 2018**
Santa Clara University

- Undergraduate research award for women in STEM, covered housing costs for summer internship at NASA Ames

SELECT PUBLICATIONS

- *Clusters, Clumps, Dust, and Gas (CCDG) in NGC 1614: Benchmarking Cluster Demographics in Extreme Systems*, **Caputo M.** et al. (2024) *AJ* 168, 259.
- *Physics and Chemistry of Radiation Driven Cloud Evolution. [C II] Kinematics of IC 59, and IC 63*, **Caputo M.** et al. (2023) *AJ* 50, 140.
- *The PHANGS-HST-Hα Survey: Warm Ionized Gas Physics at High Angular Resolution in Nearby Galaxies with the Hubble Space Telescope*, Chandar, R., Barnes, A. T., Thilker, D. A., **Caputo M.**, et al. (2025) *AJ* 169, 150.
- *A Tale of Three Dwarfs: Cluster-based Star Formation Histories of Blue Compact Dwarf Galaxies*, Chandar, R., **Caputo M.**, et al. (2024) *ApJ* 965, 95